

Proxy Card

Enliven Therapeutics, Inc.

Annual Meeting of Stockholders

For Stockholders of Record as of March 16, 2026

Thursday, May 14, 2026 – 10:00 AM Eastern Time

Held via the Internet — please visit www.proxydocs.com/ELVN

The undersigned hereby appoints the Named Proxies, and each of them, with full power of substitution, as lawful agents and proxies to vote all shares of Enliven Therapeutics, Inc. common stock that the undersigned is entitled to vote at the 2026 Annual Meeting of Stockholders, and any adjournment or postponement thereof, upon the matters set forth on the reverse side.

This proxy, if signed, dated and returned, will be voted as directed. If signed and returned without direction, the shares will be voted in accordance with the Board of Directors' recommendations shown on the reverse side. The undersigned acknowledges receipt of the Notice of Annual Meeting, Proxy Statement, and Annual Report.

Your vote matters!

Have your ballot ready and use one of the methods below:

Internet: www.proxypush.com/ELVN

Phone: 1-866-829-5209

Mail: return this card in the postage-paid envelope provided

YOUR VOTE IS IMPORTANT! Please vote by May 12, 2026, 11:59 PM ET.

Your control number

0022 1793 3306

Have the 12-digit control number in the box above available when you vote online or by phone.

Please make your marks like this: ☒ Use dark black pencil or pen only

THE BOARD OF DIRECTORS RECOMMENDS THAT YOU VOTE FOR EACH OF THE DIRECTOR NOMINEES (PROPOSAL 1), FOR PROPOSALS 2 AND 3.

PROPOSAL	YOUR VOTE			BOARD OF DIRECTORS RECOMMENDS
				↓
1. Election of Directors				
	FOR	AGAINST	ABSTAIN	
1.01 Minnie Cartwright	☐	☐	☐	FOR
1.02 Miles Leuschke	☐	☐	☐	FOR
1.03 Mike Brekke	☐	☐	☐	FOR
	FOR	AGAINST	ABSTAIN	
2. Ratification of Independent Registered Public Accounting Firm	☐	☐	☐	FOR
3. Advisory Vote on Executive Compensation	☐	☐	☐	FOR
4. Advisory Vote on Frequency of Executive Compensation Vote	☐	☐	☐	1 YEAR

*If any other matters properly come before the meeting, shares represented by properly submitted proxies will be voted on such matters in the discretion of the Named Proxies.

Only stockholders of record as of the close of business on March 16, 2026, the record date for the meeting, are entitled to receive notice of, and to vote at, the meeting.

Authorized Signatures - Must be completed for your instructions to be executed.
Please sign exactly as your name(s) appears on your account. If held in joint tenancy, all persons should sign. Trustees, administrators, etc., should include title and authority. Corporations should provide full name of corporation and title of authorized officer signing the proxy card and/or voting instruction form.